ADRIEN SEMIK LE DU DA SILVA

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EDUCATION

Non-Degree Exchange / Single Course Unit September 2026 - June 2027
Sorbonne University - Pierre and Marie Curie Campus
Quantum Mechanics - Theoretical Physics
Relevant Coursework: Foundations of Quantum Mechanics, Lie Groups and Algebras, Symmetry and Invariance Analysis, Introduction to Classical Fields and Fundamental Equations of Theoretical Physics

Master's Degree in Aerospace Engineering September 2022 - June 2027
IPSA - Institute of Advanced Technologies (Institut Polytechnique des Sciences Avancées)
Aerospace Signals and Systems
Relevant Coursework: Linear Algebra, Data Analysis, Statistical Modeling, Control Theory, Signal Processing, Electromagnetism, Wave Physics, Plasma Physics

Academic Exchange Program in Malaysia September 2025 - February 2026
Universiti Teknologi Malaysia (UTM)
Relevant Coursework: Semiconductor Physics, Numerical Methods, Machine Learning

High School Diploma (Baccalauréat) 2019 - 2022
Lycée François Mansart ; Saint-Maur-des-Fossés

SKILLS AND INTERESTS

Interests	Theoretical Mathematics & Physics, Quantum Mechanics, Calisthenics, Climbing, Soccer
Tools / Prog.	Python, C/C++, LaTeX, HTML, CSS, MATLAB, Ansys STK
Languages	French (Bilingual), Portuguese (Bilingual), English (TOEIC 875)

PROJECTS

Creation of a Scientific Article Website (Math & Physics) September 2025

- Authored notes and exercises in Quantum Mechanics and Differential Geometry
- Developed mathematical proofs and solved complex analytical problems
- Published research work on a personal academic website
- Link: https://waayyyy.github.io/My-project-in-Physics/

Design & Optimisation of Plasmonic Waveguides Sept 2026 - Jan 2027

Modeled plasmonic interactions in nanoparticle chains using dipole approximation and Green's functions

- Analyzed optical signal transmission, pulse propagation, and geometrical parameter influences
- Conducted multiphysics simulations in COMSOL for aerospace applications (sensors, optical comms)
- Developed AI-based predictive models to optimize structural configurations and system performance
- Contact supervisor : Hela.friha@ipsa.fr

EXPERIENCE

Engineering Intern, IAG - USP-University of São Paulo - Brazil June 2026 - October 2026
Institute of Astronomy, Geophysics and Atmospheric Sciences

- Design of a periodic controller combining
  an Extended State Observer (ESO)
  and a Sliding Mode Control (SMC) strategy with torque limitation limits.
- Simulation study and structural optimization
  of a rotator driven by motors and rollers.
- Under the supervision of Beatriz Barbuy for the manufacturing
  of a component for the MOSAIC telescope.

Waiter (Summer Job) June 2025 - September 2025
Restaurant in Paris

- Dining room setup and preparation for service operations.
- Welcomed international clientele and served food and beverages.
- Table setting, general cleaning, and maintenance of hygiene standards.
- Direct coordination between kitchen and bar staff.

Engineering Intern (Operations), SIAAP June 2024 - July 2024
Paris Metropolitan Sanitation Authority

- Installation of fire safety systems and analysis of job sites according to specifications.
- Isolation of multiple wastewater network valves.
- Safety awareness training for network access.